Variation of Parameters

The Wronskian

1 For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.

2 For $y_1 = \cos x$, $y_2 = \sin x$, compute W.

The variation of parameters formula

3 State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W.

A forcing term with no undetermined-coefficients trial form

4 Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.

A repeated root, with forcing dividing by x

5 Solve $y'' - 2y' + y = \dfrac{e^x}{x}$ $(x > 0)$ using variation of parameters.

Comparing with undetermined coefficients

6 Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.